Dual-Cohort APCD Pipeline Architecture: Opioid ED vs. Polypharmacy ED

Virginia All-Payer Claims Database (APCD) Data Lake (2016–2020)

Medical Claims
- Statewide Hospital & Community EDs
- 10 Diagnostic Positions (ICD-10)
- Procedure Codes (CPT/HCPCS)

Pharmacy Dispensing Claims
- Paid & Adjudicated Fills (Possession Confirmed)
- National Drug Codes (NDC) mapped to RxNorm
- Days Supply, Metric Quantity, Fill Dates

Demographic & Temporal Partitioning Engine
- 4 Study Years: 2016, 2017, 2018 (Dev/CV) | 2019 (Prospective Holdout)
- 8 Age Bands: 0–12, 13–24, 25–44, 45–54, 55–64, 65–74, 75–84, 85–114

Non-Opioid HCG Scan

Opioid Criteria Scan

Dual Mutually Exclusive Target Derivation

Branch B: Polypharmacy Adverse Drug Events (ed_non_opioid)

Clinical Target Anchor (t_0) & Cleanse
- Milliman HCG Detail Codes: P51b, O11, P33
- ABSOLUTE Opioid Cleanse: Zero opioid ICD codes across all 10 claim positions in study year

Raw ED Visits

Sequential ADE Filtering Funnel
1. Utilization Cap: < 7 ED visits/yr (Drops 'flyers')
2. Proximal Rx Window: 1–21 days pre-index (Validated by FDA FAERS: captures ~90.5% ADEs)
3. 0-Day Discharge Guard: Purges Day 0 discharge meds

True ADE Encounters

21–30 Day Acute Lookback Horizon
[t_0 - 21/30 days → t_0 - 1 day]
- Multi-Drug Concurrency & Drug-Drug Interactions
- Beers Criteria High-Risk Meds in Older Adults
- Acute Anticholinergic & Sedative Burden

Branch A: Opioid-Related ED Cohort (opioid_ed)

Clinical Target Anchor (t_0)
- ICD-10 F11.20 (Opioid Dependence)
- Acute Poisonings: T40.0X–T40.4X
- Scanned across all 10 Dx positions

Index Encounter

12-Month Chronic Lookback Horizon
[t_0 - 365 days → t_0 - 1 day]
- Longitudinal Opioid Titration & Cumulative MME
- Sedative Overlap (Benzodiazepines, Gabapentinoids)
- Behavioral Health & Chronic Pain Trajectories

Filter Contamination

Step 3b Temporal Reverse-Causality Purge
- Strips 348 Post-Index Treatment Features:
  - 164 OUD Meds (Suboxone, Naloxone, Vivitrol)
  - 84 Drug Screens (CPT 80307 Urine Toxicology)
  - 100 Post-Stabilization Secondary Billing Codes

Validated Polypharmacy Cases

Validated Opioid Cases

Shared 1:5 Matching Engine & Epidemiological Safeguards

1:5 Incidence-Density Matching Engine
- 5 Matched Controls per Target Case (No Replacement)
- Exact Stratification: Age Band (8 Strata), Sex, Calendar Year, Active Contact
- Pseudo-Index Date Assignment (t_pseudo = earliest non-ED medical claim)

Matched Cohorts

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Automated DuckDB QA Unit Assertions (qa_index_date_uniqueness.py)
• Asserts EXACTLY ONE index date per mi_person_key per partition
• Cell-Size Suppression Policy: n ≤ 10 masked per CMS/VHI protocols
• Control-Only Zero-Partition Safeguard for extreme age strata
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Validated Fact Table

Locked 682-Feature Pre-Index Whitelist
- Applied symmetrically across Cases and Controls
- Strict Toolbox Demarcation: Model Matrix vs. Dashboard Visuals

Export Final Parquets

Downstream Machine Learning & Translational Decision Support

Gold Cohort Fact Tables
s3://pgx-repository/cohorts/
(OPIOID_ED & ED_NON_OPIOID Parquets)

Parquet Ingestion

Aim 1 & 2: Tri-Model Gradient-Boosted Trees
- CatBoost (Symmetric/Oblivious Trees, Ordered Boosting)
- XGBoost (Deep Multi-Drug Interactions, 2nd-Order Hessian)
- XGBoost-RF (Bagged Forest Variance Reduction)
- Formal Feature Attribution (FFA) Boolean Horn Rules

Consensus-Risk Scores & Horn Rules

Aim 3: Consensus-Risk Triage Dashboard
- Serverless Microservice (pgx.jerome-dixon.io)
- Asynchronous MTM Review Queue for High-Risk Patients
- Human-in-the-Loop Clinical OODA Loop Decisional Authority